

Effects of Arginine Vasopressin on Islet Cells in Pancreatic Tissue Slices: Glucose-Dependent Modulation of IP₃ Receptor-Specific Responses

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eLife Assessment

This study presents a **useful** finding on the effects of arginine vasopressin (AVP) on islet cells in pancreatic tissue slices, using technically sophisticated spatio-temporal calcium recordings to confirm that AVP influences α and β cells differently depending on glucose concentrations. While the study's methods - particularly the calcium imaging techniques and peptide ligand design targeting V1b receptors - are strong, the reviewers were concerned about several aspects of the experimental design. However, the results on β -cell responses are **incomplete** and insufficient to support the manuscript's claims, especially due to the high variability of islet responses and lack of mechanistic and functional (hormone release) data. There are also concerns about the possibility of off-target effects and **incomplete** receptor specificity, noting that the study would have been significantly strengthened by inclusion of signaling pathway interrogation, hormone output assays, genetic validation (e.g., β cell-specific deletion of V1br), and receptor localization, although the work will still be of interest to researchers studying islet physiology in the context of health and diabetes.

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Abstract

Arginine vasopressin (AVP) is well known for regulating fluid volume, osmotic balance, and vascular tone. Its role in the regulation of pancreatic α and β cell function has been reported, yet its effects are not fully understood, particularly regarding its interaction with plasma glucose levels. The osmotic and volume challenges posed by hyper- and hypoglycaemia in

diabetes can be a significant complication of effective hormonal regulation of metabolism. In this study, we investigated the effects of AVP and synthetic peptide receptor agonists and antagonists on α and β cells in pancreatic tissue slices using live confocal Ca^{2+} imaging. Our findings demonstrate that AVP exerts glucose-dependent effects on both cell types. At low glucose concentrations, AVP, in combination with physiologically or pharmacologically increased cAMP levels, selectively activated α cells without significantly affecting β cells. In contrast, at higher glucose concentrations and pharmacologically elevated cAMP levels, physiological levels of AVP enhanced β cell activity, leading to increased Ca^{2+} oscillations and insulin release. In both cell types, AVP displayed a bell-shaped concentration dependence, with lower AVP concentrations stimulating hormone release and higher concentrations leading to diminished responses, consistent with inositol trisphosphate receptor (IP_3R) activation and inactivation properties. Furthermore, our results indicate that AVP acts primarily through V_{1b} receptors in β cells, with no involvement of V_{1a} , V_2 or oxytocin receptors. These findings provide new insights into the modulation of glucose-dependent release of pancreatic hormones by AVP in the context of changed blood osmolality due to hyper- or hypoglycemia in diabetes. Importantly, these results emphasize the potential of targeting AVP signaling pathways as a therapeutic approach in diabetes research, aiming to improve hormone regulation and nutrient homeostasis.

Highlights

- Highly spatio-temporally resolved imaging of islet Ca^{2+} oscillations on pancreatic tissue slices provides an in situ-like model for physiological and pharmacological approaches.
- Physiological glucose stimulation triggers non-linear β cell collective responses that must be taken into account when interpreting single concentration pharmacological experiments.
- In a high cAMP context, AVP acts through V_{1b} receptors on islet α and β cells, exhibiting a bell-shaped dependence driven by the activation-inactivation properties of IP_3 receptors.
- AVP modulates glucose-dependent effects on α and β cells in a physiological concentration range, in the presence of altered blood osmolality or volume due to hyperglycemia, or to the direct effects of hypoglycemia in diabetes.

Introduction

The most extensively studied physiological roles of arginine vasopressin (AVP; a.k.a. ADH) include: the regulation of body fluid volume by promoting water reabsorption in the kidneys, the maintenance of osmotic and electrolyte balance, thereby ensuring proper cellular function and overall physiological stability, and the contribution to cardiovascular homeostasis by regulating vascular tone and blood pressure (1, 2). Other actions of AVP, particularly its connection to various aspects of metabolic syndrome, are less frequently highlighted (3, 4). AVP has been described to elevate blood glucose levels through two primary mechanisms: enhancing glycogenolysis in the liver and promoting glucagon release from pancreatic islets (5, 6). However, the direct effects of AVP agonists and antagonists on pancreatic β and α cells remain contentious, with ongoing debates in the literature (7–10).

The physiology of pancreatic β cells, which secrete insulin, and α cells, which secrete glucagon, are intricate and remain inadequately elucidated. The principal hormones released by these cell types are essential for the metabolic homeostasis of an organism, responding dynamically to nutrient availability and metabolic intermediates present in the plasma (4, 11). These responses to nutrients are further modulated by various neurohormonal signals, including those of AVP. AVP is a nonapeptide synthesized mainly in the magnocellular AVP neurons in supraoptic and paraventricular nuclei of the hypothalamus (12). Its release from the posterior pituitary into the systemic circulation is mainly stimulated by an increase in plasma osmolality, a significant depletion in blood volume, or by diverse neuronal, endocrine (reviewed in (13)) or metabolic (14) inputs to the hypothalamus. The resulting physiologically active plasma AVP concentration in rodents and humans has been reported to be in the low to mid-picomolar range (15–17). Some studies suggest that the circulating AVP and the AVP produced in the pancreas could affect insulin and glucagon secretion via paracrine signaling (18).

One significant factor contributing to the aforementioned controversies regarding the direct effect of AVP is the glucose concentration used in experimental settings. Insulin and glucagon secretions are reciprocally correlated with plasma glucose concentrations (19). While glucose alone stimulates oscillations in intracellular Ca^{2+} concentration in pancreatic β cells and causes insulin secretion (20), AVP has been reported to work as a positive modulator for glucose-stimulated insulin release (21). Notably, when AVP is administered intraperitoneally alongside glucose in fasted mice, there is no significant increase in overall insulin secretory activity compared to glucose alone; however, it does result in a marked reduction in blood glucose levels (8). Independently of glucose levels, multiple *in vivo* studies across various species utilizing intravenous or intramuscular injections of AVP at a wide range of concentrations have demonstrated an increase in the release of pancreatic hormones (6, 14, 21–23). On the contrary, some studies have reported no or selective effects on glucagon (24, 25) and insulin release (6, 26), respectively, and there is also a report from *in vitro* experiments in rats showing AVP-dependent inhibition of insulin release (26). In addition to its effects on insulin secretion, AVP significantly enhances the proliferation of both rodent and human β cells and protects against cytokine-induced β cells apoptosis (8).

In an organism, AVP exerts its diverse physiological actions by binding to four distinct subtypes of transmembrane G-protein-coupled receptors (GPCRs): V_{1a} , V_{1b} , V_2 , as well as oxytocin receptors (27). These receptors differ in their tissue expression, signaling mechanisms, and physiological functions. V_{1a} receptors are primarily expressed in vascular smooth muscle, the heart, liver, and central nervous system, while V_{1b} receptors are predominantly found in the anterior pituitary gland and pancreas. The activation of V_{1a} receptors is associated with vascular contraction, heart function (14) and modulation of glucose and lipid metabolism, while the activation of V_{1b} receptors is involved in the release of adrenocorticotrophic hormone (ACTH), glucagon and insulin. These two types of receptors effectively couple to the $G_{q/11}$ protein and activate the phospholipase C (PLC) pathway, leading to an increase in intracellular Ca^{2+} levels, which mediates described physiological responses. In contrast to V_{1a} and V_{1b} , V_2 receptors preferentially couple to the G_s protein thereby activate the adenylate cyclase pathway and increase cyclic AMP (cAMP) levels. This signaling cascade is crucial for the regulation of water reabsorption in the kidneys as V_2 receptors are primarily expressed in the renal collecting ducts and vascular endothelial cells, facilitating the insertion of vesicles containing aquaporin channels into the cell membrane, thereby enhancing water permeability (28).

The binding of AVP to either V_{1a} or V_{1b} receptor and activation of PLC leads to intracellular Ca^{2+} release due to inositol trisphosphate (IP_3) production. Such IP_3 -mediated Ca^{2+} release has been shown to stimulate the secretion of glucagon (14) or insulin in primary (21) and clonal β cells (29). The role of $G_{q/11}$ coupled receptors has also been corroborated in pharmacological studies, where specific V_{1a} (SR-49059) or V_{1b} (Nelivaptan) receptor antagonists completely annulled, and oxytocin receptor antagonist (L-371,257) partially abolished an insulinotropic action of AVP in

rodent BRIN BD11 β cells (8 [↗](#)). Despite the fact that a plethora of receptors have been described to play a role, the majority of the literature has attributed the dominant effects of AVP on α (14 [↗](#)) and β (7 [↗](#)) cells in pancreatic islets to be mediated by the V_{1b} receptors. Recent transcriptomic investigations have revealed that V_{1b} receptors are selectively expressed on α cells, with some studies suggesting their potential use as a diagnostic marker for α cell identification (30 [↗](#)). Furthermore, research indicates that V_{1b} receptors exhibit significant enrichment, specifically within α cells (14 [↗](#)). The potential involvement of V_{1a} receptors in α cells has also not been completely ruled out (8 [↗](#)). Studies using genetically modified rodents have provided further evidence for the role of AVP and V_{1b} receptors in islet function. Mice deficient in V_{1b} receptors could not release insulin after AVP stimulation (7 [↗](#), 31 [↗](#)), with no difference in fasting blood glucose levels (7 [↗](#)). Similarly, glucagon levels were significantly reduced in $V_{1b}^{-/-}$ mice (9 [↗](#)). On the other hand, in Brattleboro rats, which carry a naturally occurring genetic mutation that completely abolishes the ability to produce AVP, have been reported to have improved glucose tolerance (32 [↗](#)).

It is important to emphasize that AVP does not act primarily as an initiator of hormone release, but rather has a permissive and potentiating role. For instance, AVP has been well-documented to facilitate the corticotropin-releasing hormone (CRH) (33 [↗](#)) or histamine-dependent (34 [↗](#)) release of adrenocorticotrophic hormone (ACTH) from the anterior pituitary, both processes that are G_s /cAMP-dependent. Both β and α cells utilize critical signaling pathways involving G_s -coupled protein receptors, where the permissive and potentiating roles of AVP may be significant (35 [↗](#), 36 [↗](#)).

And finally, given that the end readout of V_{1b} receptor activation is the activity of IP_3 receptors with well-described activation and inactivation properties (37 [↗](#)), such an arrangement could accommodate the broad spectrum of previously observed effects of AVP on the function of α and β cells. On that note, the present study reassessed the effects of AVP and the selection of specific novel AVP receptor agonists and antagonists on pancreatic β and α cells using fresh pancreatic tissue slices combined with live confocal imaging of intracellular Ca^{2+} oscillations. In the tissue slice, the majority of AVP stimulatory activity occurs within the known physiological AVP range (1 [↗](#)), a range that correlates with the correction of plasma osmolarity involving water retention in the distal nephron or oral water intake, and all physiological scenarios related to blood volume depletion.

Material and methods

Ethics Statement

The study strictly adhered to all national and European guidelines regarding the care and treatment of laboratory animals. Every effort was made to minimize animal suffering and improve animal welfare. The experimental protocol received approval from the Administration of the Republic of Slovenia for Food Safety, Veterinary and Plant Protection (grant No.: U34401-12/2018/2) and The Ministry of Education, Science and Research, the Republic of Austria (GZ 2022-0.325.009).

Animals, Tissue Slice Preparation and Indicator Loading

Experiments were conducted in 40, 8–25-week-old C57BL/6J mice of both sexes, which were housed in individually ventilated cages (Allentown LLC, USA) at a room temperature 22–24 °C and 45–55% relative humidity with a 12-hour light/dark cycle. The mice had *ad libitum* access to water and a standard chow (Ssniff, Soest, Germany). Acute pancreatic tissue slices were prepared as previously described (38 [↗](#)). Briefly, after CO_2 anesthesia, cervical dislocation and opening of the abdominal cavity, the common bile duct was clamped distally at the major duodenal papilla 1.9% low melting point agarose (Lonza, Basel, Switzerland) was injected proximally. Agarose was

dissolved in an extracellular solution (ECS) containing (in mM) 125 NaCl, 26 NaHCO₃, 6 glucose, 6 lactic acid, 3 myo-inositol, 2.5 KCl, 2 Na-pyruvate, 2 CaCl₂, 1.25 NaH₂PO₄, 1 MgCl₂, 0.5 ascorbic acid and kept in a prewarmed water bath at 40 °C. Following the agarose injection, the pancreatic tissue was immediately cooled with ice-cold ECS, extracted from the abdominal cavity, and placed into a large Petri dish containing ice-cold ECS. Next, 3–5 mm³ large tissue cubes were cut from the pancreas, cleared from the connective tissue, and embedded into agarose. The embedded tissue blocks were sectioned into 140 µm thick slices using a vibratome (VT1000S, Leica Microsystems, Wetzlar, Germany). The tissue slices were stored at room temperature in HEPES-buffered saline containing 6 mM glucose (HBS, consisting of (in mM) 150 NaCl, 10 HEPES, 6 glucose, 5 KCl, 2 CaCl₂, 1 MgCl₂; titrated to pH=7.4 with 1 M NaOH). For Ca²⁺ indicator loading, the slices were incubated for 50 minutes in 3.33 ml of HEPES-buffered saline containing 6 mM glucose, 3.75 µl dimethylsulfoxide, 1.25 µl Pluronic F-127, and 6 µg Calbryte 520 AM (AAT Bioquest, Pleasanton, CA, USA). Unless specified otherwise, all chemicals were obtained from Sigma Aldrich (St. Louis, MO, USA).

Compound Synthesis, Purification, and Quality Control

VP analogs [(D-Leu)²,Ile³,Thr⁴]-VP, d[Cha⁴,Dab⁸]-VP, and d[(CH₂)₅¹,Tyr(Me)²,Dab⁵,Tyr⁹]-VP were prepared following established procedures *via* Fmoc-SPPS (9-fluorenylmethyloxycarbonyl solid-phase peptide synthesis) and standard purification and quality control methods (39) (Fig. 5, Suppl. Fig. 1, Suppl. Table 1). Shortly, peptides were obtained through manual synthesis on a Rink amide aminomethyl resin followed by TFA cleavage, oxidative folding in a 0.1 M ammonium bicarbonate buffer (pH 8.2, 24 h), and purification *via* reversed phase-high performance liquid chromatography (RP-HPLC) on a Waters Auto Purification HPLC-UV system (Kromasil Classic C₁₈ 21.2 x 250 mm, 300 Å, 10 µm) at a solvent flow of 20 mL/min and a linear gradient of 5–45% solvent B in solvent A (A: ddH₂O + 0.1 % TFA, B: ACN + 0.08 % TFA) in 50 min. Reactions were monitored by analytic RP-HPLC-mass spectrometry (RP-HPLC-MS) on a Thermo Scientific Dionex Ultimate 3000 system equipped with a Waters XSelect CSH UPLC C₁₈ XP column (3.0 x 75 mm, 130 Å, 2.5 µm), UV detection (measurement at 214 nm and 280 nm), and Thermo Scientific MSQ Plus ESI-MS unit (positive ionization mode). Compound purity was determined through analytical RP-HPLC peak integration at 214 nm, and compound identity was verified through direct injection MS (Supplementary Information). A Thermo Fisher Scientific Vanquish Horizon UHPLC system using a designated column (Kromasil Classic C₁₈ 2.1 x 100 mm, 100 Å, 5 µm) was used to determine the product concentration based on peptide absorption at 214 nm, and a comparison of absorption areas to peptide standards with known concentration (40).

Pharmacological Characterization

HEK293 cells stably expressing the EGFP-tagged receptor of interest (human oxytocin, V_{1a}, or V_{1b} receptor) were cultured. Radioligand displacement was performed in duplicates of cell membranes (6–20 µg) incubated with the radioactive agonist ³H-OT or ³H-VP and the competing peptide according to previously published protocols (41–43). The used radioligand concentrations refer to the K_d of each receptor subtype which was obtained from saturation binding experiments (0.65 nM for OTR, 0.21 nM for V_{1a}R and 0.14 nM for V_{1b}R). Nonspecific binding was determined by addition of 10 µM OT or VP. After one hour, the reaction was stopped by filtering through glass fiber filters, which were subsequently used to determine the retained radioactivity measured by liquid scintillation.

Activation of G_{q/11}-signalling was measured by IP₁ quantification using the homogeneous time-resolved fluorescence IP-One assay kit (Revvity) according to the manufacturer's recommendations. Briefly, stable cell lines were seeded at a density of 10,000 cells per well and incubated for 2 days. At the time of the assay, the cells were equilibrated in the provided stimulation buffer for 15 min prior to the addition of the peptide ligands. After 1 h at 37°C, the

	K_i ± SD (nM)			EC₅₀ ± SD (nM)	E_{max} ± SD (%)	pA₂
	OTR	V _{1a} R	V _{1b} R	V _{1b} R	V _{1b} R	V _{1a} R
OT	0.5 ± 0.2	18 ± 5	588 ± 19	n.d.	n.d.	-
AVP	8 ± 5	0.2 ± 0.1	0.3 ± 0.1	15 ± 2	97 ± 4	-
[(D-Leu) ² ,Ile ³ ,Thr ⁴]-VP	209 ± 36	5 ± 1	70 ± 7	361 ± 176	79 ± 8	n.d.
d[Cha ⁴ ,Dab ⁸]-VP	54 ± 19	23 ± 5	0.2 ± 0.02	130 ± 52	104 ± 13	n.d.
d[(CH ₂) ₅ ¹ ,Tyr(Me) ² ,Dab ⁵ ,Tyr ⁹]-VP	2300 ± 190	66 ± 15	>10,000	-	-	5.27

Cha = cyclohexylalanine; (CH₂)₅ = β-mercapto-β,β-cyclopentamethylenepropionic acid; d = deaminated N-terminus; Dab = 2,4-diaminobutyric acid.

Table 1

Affinity, potency and efficacy of [(D-Leu)²,Ile³,Thr⁴]-AVP, d[Cha⁴,Dab⁸]-AVP, and d[(CH₂)₅¹,Tyr(Me)²,Dab⁵,Tyr⁹]-AVP.

Affinity (K_i) and potency (EC₅₀) data are indicated as mean ± SD (nM, n = 3); K_i values were calculated from IC₅₀ values according to Cheng and Prusoff, assuming K_d values of 0.65 nM for OTR, 0.21 nM for V_{1a}R and 0.14 nM for V_{1b}R. Efficacy (E_{max}) data is normalized to the maximum IP₁ formation by the control. #controls were OT at the OTR or AVP at the V_{1a}R and V_{1b}R; n.d., not determined.

stimulation was stopped by adding labeled immunoreagents. The mixtures were then incubated for a further hour at room temperature before the fluorescence ratio 665/620 nm was measured on a FlexStation 3 plate reader (Molecular Devices, USA) at an excitation wavelength of 340 nm.

Data analysis of the pharmacological characterization was performed with GraphPad Prism (GraphPad Software, USA) ([Fig. 5](#), [Table 1](#)). The potency (EC_{50}) and maximum efficacy (E_{max}) values were generated by fitting the obtained data to three-parameter nonlinear regression curves with a bottom constrained to zero. Graphs were normalized to the highest response by the control, i.e. oxytocin or vasopressin. For radioligand displacement IC_{50} values were calculated by a three-parameter logistic Hill equation and then further used to approximate K_i values according to Cheng and Prusoff ([44](#)). The normalization refers to the specific binding of the radioligand in the absence of peptides as a maximum with an average of 4.2 pmol/mg for OTR, 2.6 pmol/mg for $V_{1a}R$ and 2.9 pmol/mg for $V_{1b}R$. All data regarding pharmacological characterization were presented as mean $\pm$ SD of at least three independent experiments unless otherwise stated.

Calcium Imaging

Imaging was performed with upright confocal microscope Leica TCS SP5 AOBS Tandem II (20x HCX APO L water immersion objective, NA 1.0) and inverted confocal microscope Leica TCS SP5 DMI6000 CS (20x HC PL APO water/oil immersion objective, NA 0.7). The Ca^{2+} indicator Calbryte 520 was excited using a 488 nm argon laser, and the emitted fluorescence was detected by Leica HyD hybrid detectors in photon-counting mode (Leica Microsystems, Wetzlar, Germany) in the range of 500-700 nm, as previously described ([38](#)). The laser power was adjusted to maintain a satisfactory balance between photobleaching and signal-to-noise ratio. The optical imaging thickness was set to nearly 5 μm to prevent recording from multiple cell layers. Time series were acquired with a frequency of 20 Hz and a resolution of 256 x 256 pixels and pixels size of approximately 1 μm^2 .

Stimulation Protocol

Before imaging, the stained slices were kept in a substimulatory glucose concentration (HEPES with 6 mM glucose) at room temperature. For Ca^{2+} imaging slices were transferred into the recording chamber on the microscope stage, perfused continuously with carbonated ECS containing 6 mM glucose at 37 °C. After recording basal Ca^{2+} signals in 6 mM glucose, the perfusion was exchanged with stimulatory ECS containing 8 (or 9 mM as indicated) glucose for 15-20 minutes. When a stable second phase plateau response with fast Ca^{2+} oscillations was achieved ([45](#)), AVP or AVP receptor agonist/antagonist was added to the perfusion for 10-15 minutes. Following AVP/agonist/antagonist testing, the perfusion was again changed to ECS with either 8 or 9 mM glucose for wash-out period of about 15-20 minutes and then to the substimulatory ECS with 6 mM glucose until the cessation of fast Ca^{2+} oscillations. For AVP concentration-dependent responses, islets were stimulated with 8 mM glucose and 500 nM forskolin until the plateau phase has been achieved. A version of this experiment has been performed with 2.5 nM epinephrine. In both cases, a series of AVP concentrations were applied, each of which was static incubated in the recording chamber for 15 minutes before the next concentration was applied.

Insulin measurements

For acute insulin release in response to glucose, mouse pancreatic tissue slices were preincubated (1 h) in ESC buffer containing 6 mM glucose at 37°C. The sample for the insulin release from the slices was then collected in ESC with 6 mM glucose for 5 min (basal), after which the solution was replaced with stimulatory glucose in ESC 8 mM glucose or ESC 8 mM glucose plus 10 nM AVP and incubated for 15 min. Each solution replacement step involved washing the dish containing the slices at least 3 times to ensure no remaining insulin being present. Insulin concentration was determined using the Insulin Ultra-Sensitive assay kit from Cisbio (Bagnols-sur-Ceze, France).

Data Analysis

The analysis of Ca^{2+} events has been performed as previously described (45). Briefly, the movies were processed using a custom Python script to detect ROIs corresponding to individual cells automatically. Cells have been identified based on their typical activity pattern and intraislet localization as described before (46, 47). β cells have been inactive at non-stimulatory glucose concentration (6 mM) and activated in a biphasic manner when stimulated with either 8 or 9 mM glucose. In contrast, α cells were still active at 6 mM glucose, progressively inhibited in 8 mM glucose, and further stimulated during epinephrine stimulation. Smooth muscle cells were localized around vascular structures in the slices and were used as a readout of V_{1a} receptor activity. All other cells outside the histologically identifiable islet were discarded from further analysis. In the next step, Ca^{2+} events from each ROI were automatically distilled and annotated (45).

For this study, altogether 172 pancreatic tissue slices have been imaged altogether. In addition to the frequency of events per roi per minute (epmpr), for each detected Ca^{2+} event, we measured the height, duration at half of the amplitude (halfwidth) and area under the curve (AUC) (Fig. 2E). Halfwidths between 0.01 to 100 seconds were collected and were pooled together for the global analysis. For the analysis within individual islets, the parameters from each ROI were checked for normality and then parametric analysis (ANOVA/Tukey HSD post hoc) or non-parametric analysis (Kruskal-Wallis/Dunn's test post hoc) has been performed. A similar strategy has been used when we compared several islets exposed to the same treatment. The rois with less than 5 events in the whole length of recording were eliminated from the analysis. The p-value below 0.05 has been taken as statistically significant.

Results

The glucose-dependence of AVP modulation of α and β cell activity

We have fully reproduced the previously reported glucose-dependent effects of AVP on both α and β cells (Fig. 1). It is crucial to emphasize that at the substimulatory glucose, neither forskolin, an activator of cAMP production alone nor in combination with AVP could significantly activate β cells (Fig. 1BF). On the other hand, forskolin alone or combined with AVP, activated α cells (Fig. 1DG). However, subsequent increase to stimulatory glucose in the same islet activated β cells (Fig. 1B) and progressively reduced the activity of α cells (Fig. 1D). The activation of α cells with AVP did not have a significant effect on β cells (Fig. 1BD).

As previously observed, the application of forskolin to islets after stimulation with physiological levels of glucose (8 or 9 mM), significantly increased the activity of β cells (Fig. 2B). Forskolin, even at low concentration used here (500 nM), increased the frequency of Ca^{2+} oscillations in all islets tested at 8 mM glucose (Fig. 2GE). Application of 10 nM AVP further increased the oscillation frequency, which was often associated with a visible decrease in the halfwidth of the events (Fig. 2B) and in the overall AUC, a parameter that reflects both changes in oscillation frequency and duration (Fig. 2H), although this was not significantly different after pooling the data (Fig. 2GH). It is worth noting that the AUC values varied significantly between different islets, ranging from an increase in overall activity after AVP application to no change or even a decrease in AUC.

Using low levels of forskolin in the same islet increased the activity of α cells despite the presence of high glucose concentration and glucose-dependent insulin release from β cells, both of which should inhibit the activity of α cells (Fig. 2DF). This increased activity was further enhanced by adding 10 nM AVP (Fig. 2DF). Again, the intense stimulation of α cells had no detectable effect

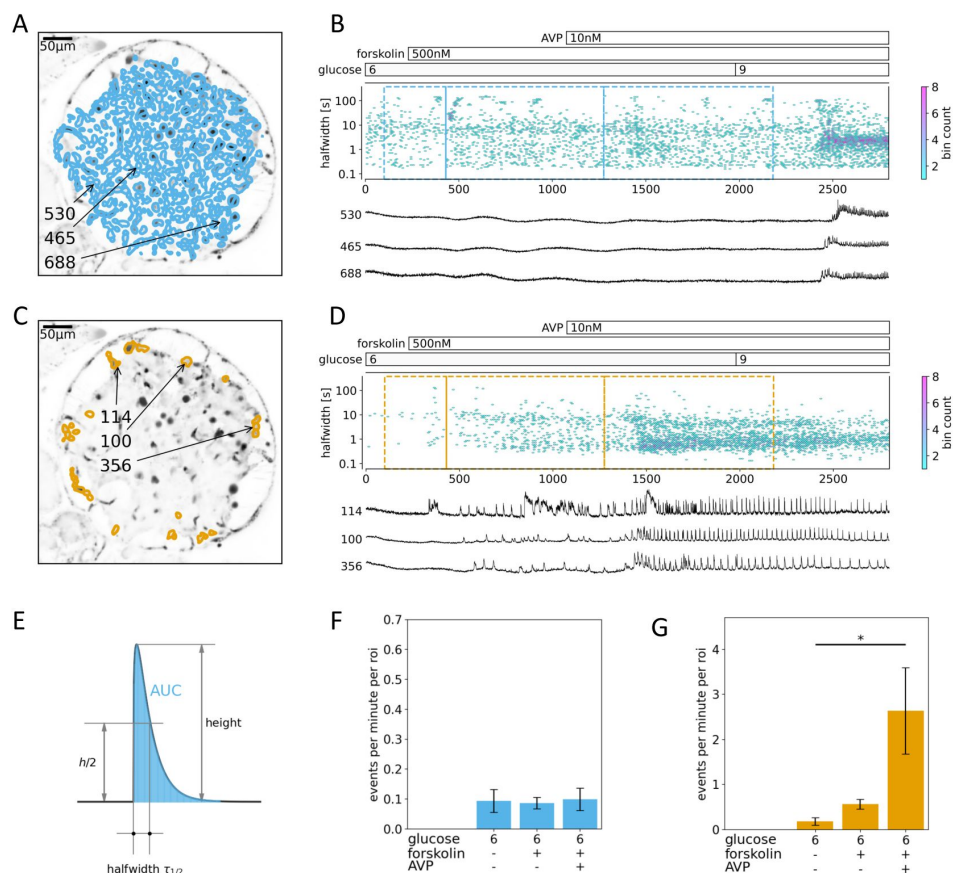


Figure 1.

The effect of AVP at the substimulatory glucose on β (A, B, F) and α (C, D, G) cells.

(A, C) Regions of interest (ROIs) were obtained with our analytical pipeline. Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B, D) (top panel) Time distribution of all measured events halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 3 time periods where events per minute per ROI parameter was sampled for panels F and G. (bottom panel) Representative traces from ROIs indicated in A and C. (E) A scheme showing how the height, duration at half amplitude (halfwidth) and area under the curve (AUC) were measured for each Ca^{2+} event detected. (F, G) Events per minute per ROI were normally distributed, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD. The significant differences in AVP treatment at the substimulatory glucose were found in α cells only (* $p < 0.05$).

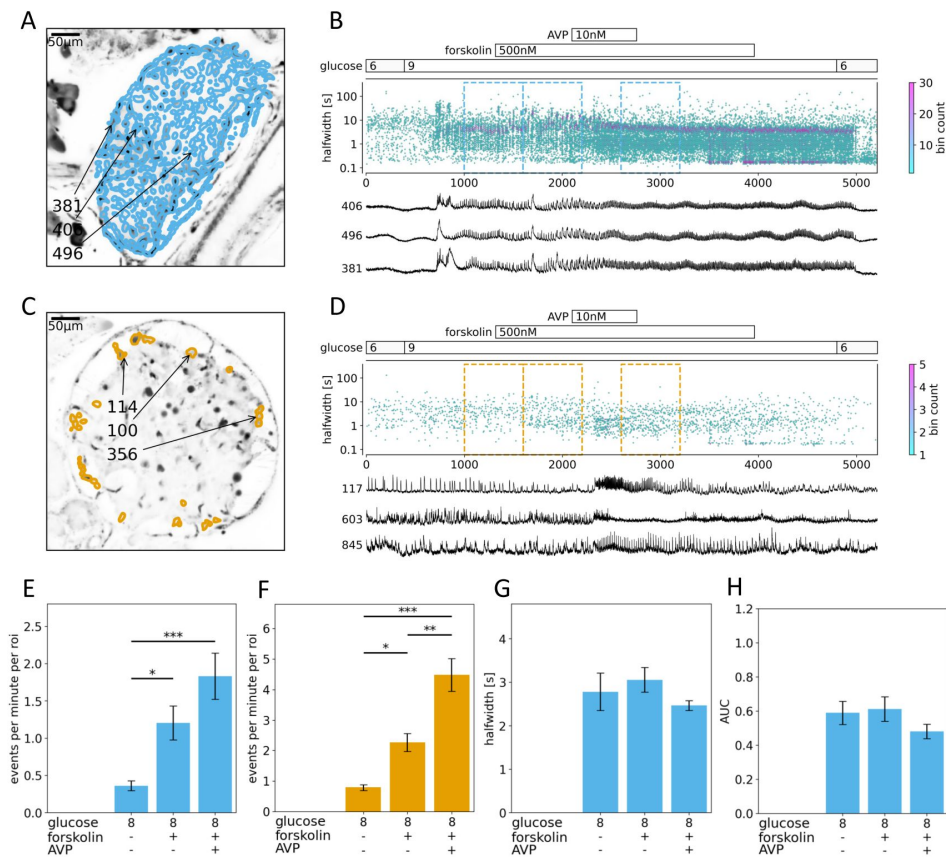


Figure 2.

The effect of AVP in the physiological stimulatory glucose range on β (A, B, E, G, H) and α (C, D, F) cells.

(A, C) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B, D) (top panel) Time distribution of all measured events' halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 3 time periods where events per minute per ROI, Halfwidth and AUC parameters were sampled for panels E-H. (bottom panel) Representative traces from ROIs indicated in A and C. (E, F) Forskolin significantly increased the number of Ca^{2+} oscillations per minute per ROI in β cells that were further stimulated by addition of AVP. A similar effect was observed also in α cells. (G, H) A non-significant but visible decrease in the events' halfwidth and AUC values due to forskolin and AVP administration in β cells were noted. The parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

on AVP modulation of β cell activity. The control experiment without AVP application did not produce significant changes in oscillation frequency in a comparable recording time (**Suppl. Fig. 3**). In this study we did not observe any sex differences in AVP responses.

The effect of physiological epinephrine concentration on AVP modulation of α and β cells

To further corroborate the permissive role of AVP on cytosolic cAMP levels we used physiological epinephrine concentrations. We have previously shown (47) that epinephrine can reduce cAMP concentration and suppress Ca^{2+} activity in β cells treated with physiological glucose concentration through G_i -protein coupled pathway, epinephrine can reduce cAMP concentration and suppress stimulated by the physiological glucose concentration, while simultaneously increasing cAMP production in α cells through the G_s -protein coupled pathway.

In the stimulatory 9 mM glucose, the administration of epinephrine rapidly reduced the frequency of oscillations in β cells to a resting activity level (**Fig. 3**). Subsequent perfusion with progressively increasing AVP concentration did not recover the epinephrine inhibition beyond this resting activity level, indicating that the AVP effect in β cells does not involve activation of G_s -protein coupled V_2 receptors (**Fig. 3E**).

On the other hand, α cells were prominently activated by AVP and responded in a frequency-coded epinephrine concentration-dependent manner (**Fig. 3DF**). This provided further evidence for the permissive and potentiating roles of AVP. The use of forskolin therefore allowed us to study the AVP dependence of both α and β cells at the same time. Epinephrine experiments were performed without forskolin, further demonstrating that its addition was not necessary for the activation of both α and β cells, but solely to support the activation of AVP receptors.

AVP enhances the activity of α and β cell within its physiological osmoregulatory range

To further investigate the sources of variability in the observed AVP action in the presence of forskolin and 8 mM glucose, we performed a series of experiments in which slices were exposed to an AVP concentration ramp. AVP administration reproducibly induced concentration-dependent activation of both α and β cells (**Fig. 4**). The most prominent stimulatory effect of AVP on both cell types was observed in its physiological osmoregulatory range between 10 and 100 pM (**Fig. 4FG**). In β cells, lower concentrations of AVP resulted in an increased frequency of Ca^{2+} oscillations, but this was reduced at higher AVP concentrations (**Fig. 4G**), whereas the halfwidth of oscillations decreased progressively and significantly with increasing AVP concentration (**Fig. 4J**). The overall effect of AVP produced a bell-shaped distribution of both the frequency parameter, measured as events per minute per region of interest (epmpr) (**Fig. 4F**), and AUC values (**Fig. 4G**). Such bell-shaped relationships align with the activation and inactivation properties of IP_3 receptors (37), which are the primary target signaling pathway of AVP stimulation in these cells. The IP_3 receptors are activated with lower IP_3 and Ca^{2+} levels, but become inactivated at higher concentrations. Unphysiologically high levels of AVP lead to strong stimulation of G_q protein-coupled receptors, contributing to the inactivation of IP_3 receptors and effectively diminishing the dynamic range of the AVP response, potentially resulting in an inhibitory effect with AUCs below that of 8 mM glucose stimulation alone (**Fig. 4G**). This inhibitory effect was also reflected in the reduced insulin release (**Fig. 4E**). Such a paradoxical inhibitory effect on β cells has previously been documented in relation to muscarinic receptor signaling, which similarly involves the activation of IP_3 receptors (48). Notably, peak AVP activation in α cells occurs at an order of magnitude higher AVP concentration (**Fig. 4G**), exhibiting limited inactivation at physiological AVP levels, which supports previously published findings (14). Additionally, at supraphysiological glucose concentration, the AVP-dependent

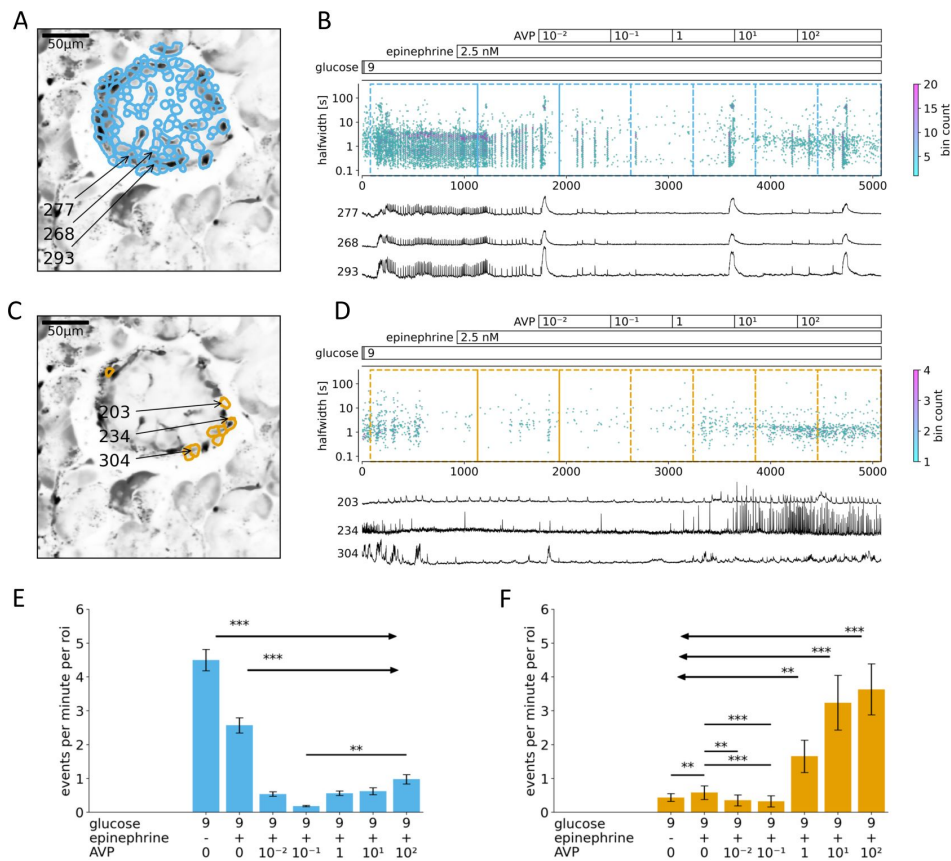


Figure 3.

The effect of physiological epinephrine concentration β (A, B, E) and α (C, D, F) cells.

(A, C) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B, D) (top panel) Time distribution of all measured events halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 7 time periods where events per minute parameter was sampled for panels E and F. (bottom panel) Representative traces from ROIs indicated in A and C. (E, F) The analysis of events per minute per ROIs shows that inhibition by epinephrine of glucose-dependent activation is reversed with increasing vasopressin concentration (0.01, 0.1, 1, 10 and 100 nM) in α cells but not in β cells. The parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD (* p < 0.05, ** p < 0.01, *** p < 0.001).

activation/inactivation curve for β cells was shifted to the left (Fig. 4I), allowing for inhibitory modulation by AVP at lower concentrations and suggesting that glucose sensitivity is driving sensitivity to AVP.

V_{1b} receptor agonist modulation of α and β cell activity

One of the primary objectives of this study was to evaluate the exclusive role of V_{1b} receptors without confounding effects of the activation of V_{1a}, V₂ and oxytocin receptors in α and β cells using fresh pancreatic tissue slices. To accomplish this, we employed AVP, alongside a set of newly synthesized specific peptides (Fig. 5A): d[Cha⁴,Dab⁸]-VP, a specific V_{1b} receptor agonist; [(D-Leu)²,Ile³,Thr⁴]-VP, a mixed V_{1b} receptor agonist and V_{1a} and oxytocin receptor antagonist; and d[(CH₂)₅¹,Tyr(Me)²,Dab⁵,Tyr⁹]-VP, specific V_{1a} receptor antagonist. Further, we used a commercially available compound tolaptan to further substantiate the absence of V₂ receptors in β cells – evidenced by our inability to reverse epinephrine inhibition with AVP (Fig. 3).

The three peptide ligands were rationally designed based on available earlier structure-activity relationship data from our own previous work (43) and others (49–51), incorporating specific modifications to optimize receptor selectivity and activation. The functional properties of the three peptide ligands were assessed using a panel of cell-based *in vitro* assays expressing human oxytocin, V_{1a}, and V_{1b} receptors. Receptor-subtype selectivity was assessed using radioligand binding studies (Fig. 5B) measuring the displacement of tritiated OT/AVP with increasing concentrations of peptides. This demonstrated that d[Cha⁴,Dab⁸]-VP had the highest affinity for V_{1b} receptor (200 pM) and a selectivity gain over oxytocin (~270-fold) and V_{1a} receptor (~115-fold) (Table 1). On the other hand, [(D-Leu)²,Ile³,Thr⁴]-VP had comparable affinity for all three receptors, ranging between 5 nM and 209 nM (Table 1), and lastly d[(CH₂)₅¹,Tyr(Me)²,Dab⁵,Tyr⁹]-VP exhibited only affinity for V_{1a} receptor (66 nM), but did not bind to oxytocin (>2.3 μ M) or V_{1b} receptors (>10 μ M) (Table 1).

To determine their activation profile, intracellular signaling was evaluated using IP1 second messenger quantification in two steps: a first single-concentration screen allowed to distinguish agonist/antagonist properties of the peptide ligands (Fig. 5C) and a comprehensive concentration-response determination yielded reliable values for potency (EC₅₀), efficacy (E_{max}) and inhibitory potency (pA₂) (Fig. 5DE). Peptide d[Cha⁴,Dab⁸]-VP is a moderate potent full agonist at V_{1b} receptor (EC₅₀ = 130 nM; E_{max} = 104%) (Table 1), with no significant activation of V_{1a} or oxytocin receptors (Fig. 5C). Its potency was slightly reduced as compared to endogenous AVP (15 nM). Peptide [(D-Leu)²,Ile³,Thr⁴]-VP functioned as a moderate/weak partial agonist at V_{1b} receptor (EC₅₀ = 361 nM; E_{max} = 79%) (Table 1), while displaying antagonist properties at oxytocin and V_{1a} receptors (Fig. 5C). Peptide d[(CH₂)₅¹,Tyr(Me)²,Dab⁵,Tyr⁹]-VP demonstrated selective antagonism for V_{1a} receptor, effectively inhibiting endogenous ligand-induced signaling in a competitive manner (as exemplified by Schild regression analysis, Fig. 5D). Its inhibitory potency (pA₂) was determined to 5.27 (Table 1), corresponding to 5.37 μ M. These findings demonstrate that the newly designed peptide ligands provide valuable tools for probing the role of V_{1b} receptors in α and β cells.

Applied in pancreatic tissue slices, [(D-Leu)²,Ile³,Thr⁴]-VP demonstrated an effect similar to AVP, characterized by an increase in the frequency and a decrease in the halfwidth of the Ca²⁺ oscillations (Fig. 6). However, some islets exhibited no change or a reduction in Ca²⁺ oscillations frequency as shown by paired analysis (before-after AVP application), resulting in no significant difference in the pooled AUC data (Fig. 6E). At the same time α cells have been activated in a frequency-coded manner (Fig. 6DF).

Insulin release from islets within pancreas slices did not show statistically significant differences following treatment with [(D-Leu)²,Ile³,Thr⁴]-VP in 8 mM glucose and substimulatory concentration of forskolin (Fig. 6DG). Given that [(D-Leu)²,Ile³,Thr⁴]-VP also antagonized V_{1a}

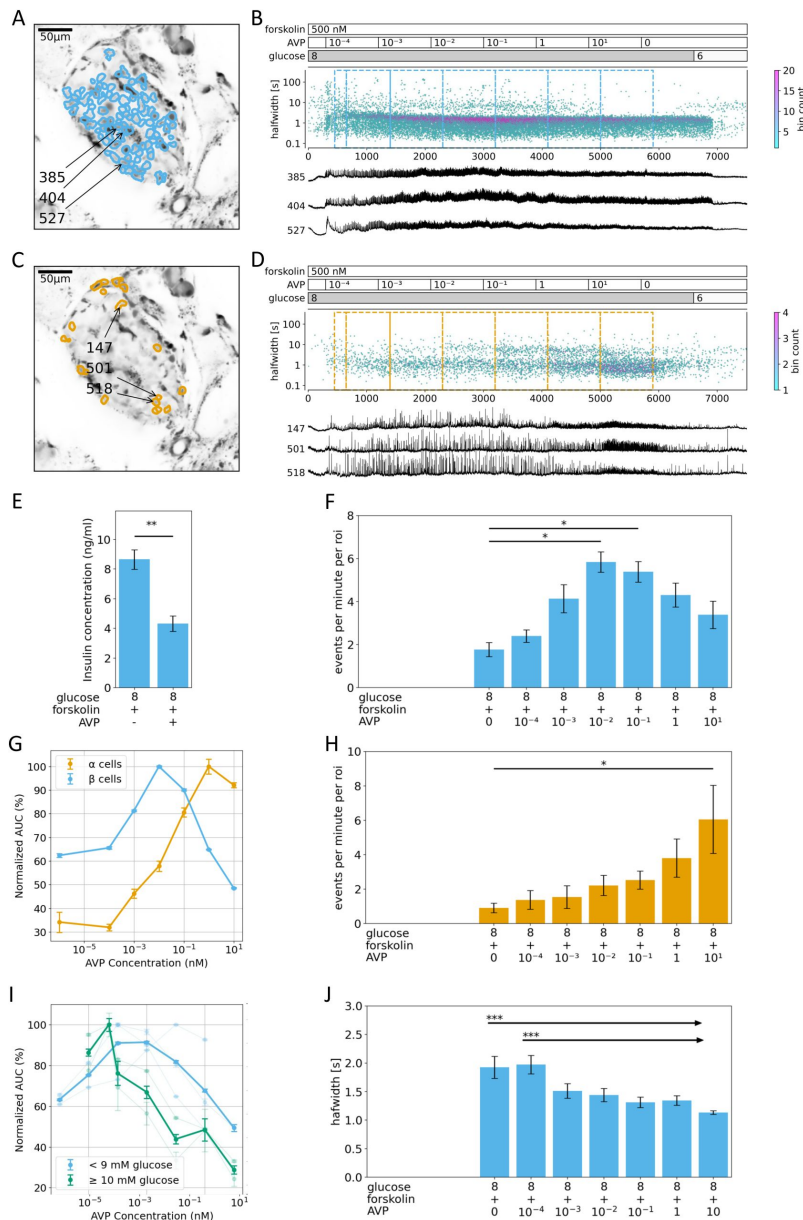


Figure 4.

AVP modulates the activity of β (A, B, E-G, I, J) and α (C, D, G, H) cells in a concentration-dependent manner.

(A, C) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B, D) (top panel) Time distribution of all measured events halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 7 time periods where events per minute per ROI and halfwidth parameters were sampled for panels F, H and J. (bottom panel) Representative traces from ROIs indicated in A and C. (E) Supraphysiological concentrations of AVP (above 10 nM) reduced insulin release from β cells below that measured at 8 mM glucose and forskolin alone. (F, H) Events per minute per ROI in β cells was gradually increased from 0.0001 nM to the physiological osmoregulatory range of AVP, between 0.01 and 0.1 nM, and then gradually decreased with further increases in AVP concentration. In the same islets the number of events per minute increased gradually in α cells also at higher AVP. (G) Pooling several islets revealed that bell-shaped distribution of the AUC could be found in both β and α cells, although the latter being shifted towards higher AVP concentration. (I) The AVP-dependent activation/inactivation curve for β cells was left-shifted at supraphysiological glucose concentration. (J) The halfwidth of events in β cells progressively declined with increased AVP concentration. All the parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD (* p < 0.05, ** p < 0.01, *** p < 0.001).

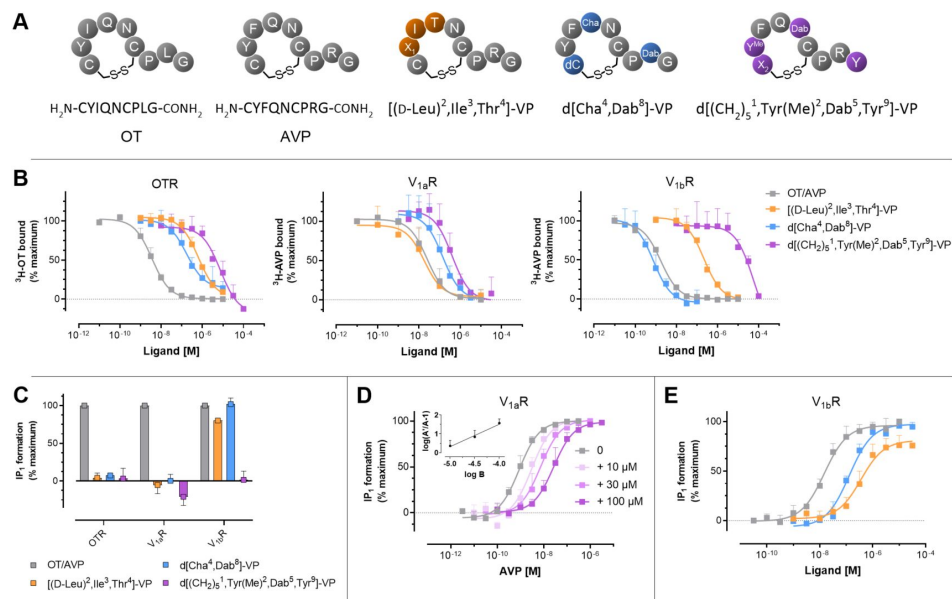


Figure 5

Schematic structures and pharmacology of peptide probes $[(\text{D-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$, $\text{d}[\text{Cha}^4, \text{Dab}^8]\text{-VP}$, and $\text{d}[(\text{CH}_2)_5^1, \text{Tyr}(\text{Me})^2, \text{Dab}^5, \text{Tyr}^9]\text{-VP}$.

(A) Schematic structures and amino acid compositions of oxytocin (OT), vasopressin (AVP), and their derivatives. **(B)** Radioligand displacement assay of $[(\text{D-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$, $\text{d}[\text{Cha}^4, \text{Dab}^8]\text{-VP}$, and $\text{d}[(\text{CH}_2)_5^1, \text{Tyr}(\text{Me})^2, \text{Dab}^5, \text{Tyr}^9]\text{-VP}$. Concentration-dependent displacement of ^3H -OT (0.65 nM) or ^3H -AVP (0.21 nM for $\text{V}_{1\text{aR}}$, 0.14 nM for $\text{V}_{1\text{bR}}$) in HEK293 cell membranes expressing the human OTR, $\text{V}_{1\text{aR}}$ or $\text{V}_{1\text{bR}}$ by either test ligand or the respective control. Data is presented as specific binding by subtracting nonspecific from total binding normalized to the maximum binding of the radioligand in the absence of the peptides ($n = 3$). **(C)** Ligand induced formation of intracellular IP_1 by 10 μM OT/AVP, $[(\text{D-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$, $\text{d}[\text{Cha}^4, \text{Dab}^8]\text{-VP}$, and $\text{d}[(\text{CH}_2)_5^1, \text{Tyr}(\text{Me})^2, \text{Dab}^5, \text{Tyr}^9]\text{-VP}$ demonstrating agonistic properties of $[(\text{D-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$ and $\text{d}[\text{Cha}^4, \text{Dab}^8]\text{-VP}$ at $\text{V}_{1\text{bR}}$, whereas in the sense of an antagonism no increased IP_1 accumulation was observed at OTR and $\text{V}_{1\text{aR}}$ ($n = 2$). Activation data was normalized to maximum response generated by the control (OT for OTR, AVP for $\text{V}_{1\text{aR}}$ and $\text{V}_{1\text{bR}}$). All data points are shown as mean $\pm$ SD. **(D)** Antagonistic properties of $\text{d}[(\text{CH}_2)_5^1, \text{Tyr}(\text{Me})^2, \text{Dab}^5, \text{Tyr}^9]\text{-VP}$ at $\text{V}_{1\text{aR}}$ were assessed through concentration-dependent displacement of AVP according to the Schild method ($n=3$). **(E)** Concentration-response curves of AVP, $[(\text{D-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$, and $\text{d}[\text{Cha}^4, \text{Dab}^8]\text{-VP}$ were obtained for the $\text{V}_{1\text{bR}}$ in order to assess their potency ($n = 3$). Affinity constants (K_i), potency (EC_{50}) and maximum efficacy (E_{max}) values for $[(\text{D-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$, $\text{d}[\text{Cha}^4, \text{Dab}^8]\text{-VP}$, $\text{d}[(\text{CH}_2)_5^1, \text{Tyr}(\text{Me})^2, \text{Dab}^5, \text{Tyr}^9]\text{-VP}$, and the respective controls are listed in Table 1. Abbreviations: Cha = cyclohexylalanine; Dab = 2,4-diaminobutyric acid; d = desamino; X_1 = D-Leu; X_2 = $\text{d}(\text{CH}_2)_5$ (β -mercapto- β , β -cyclopentamethylenepropionic acid); Y^{Me} and $\text{Tyr}(\text{Me})$ = O-methylated tyrosine.

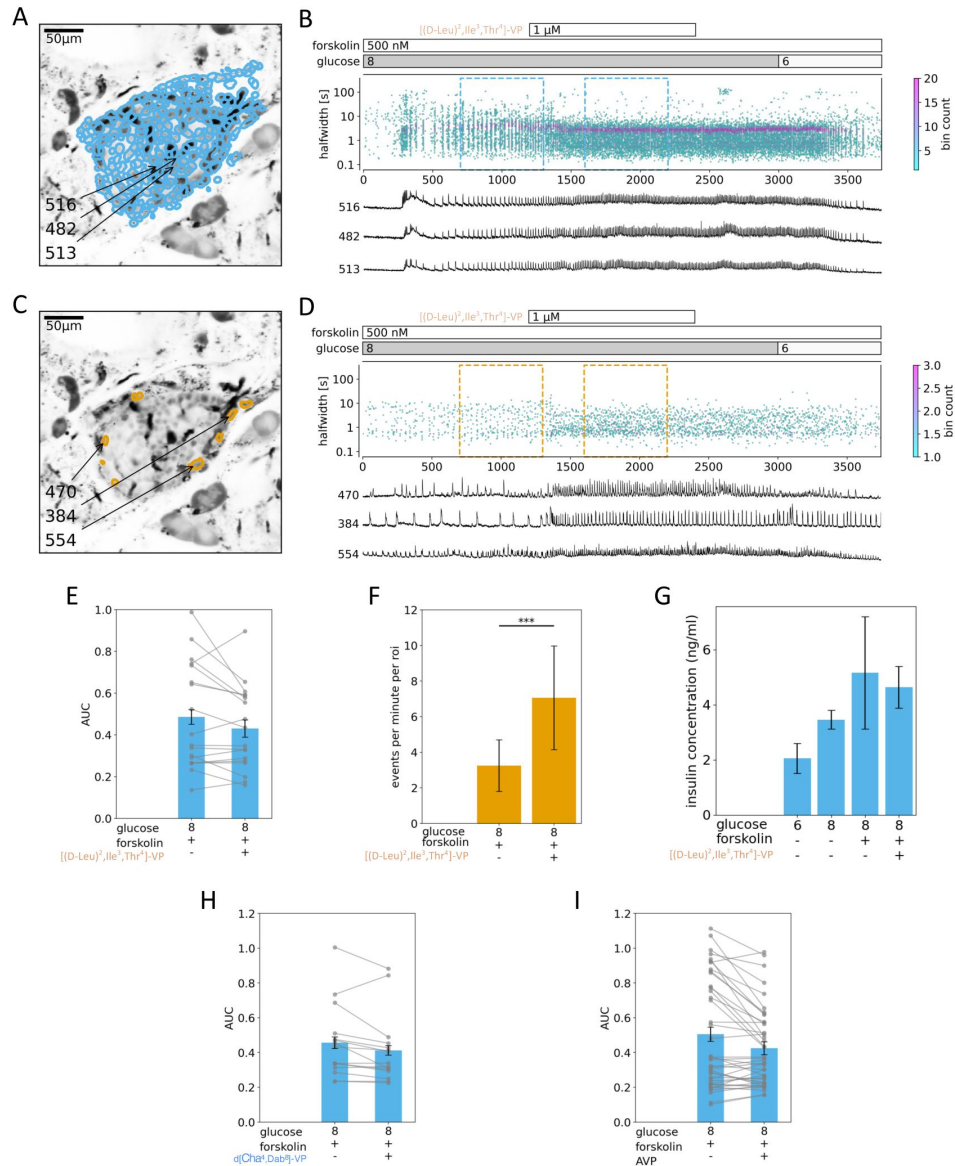


Figure 6.

A specific V_{1b} receptor agonist and V_{1a} and oxytocin receptor antagonist, has a similar effect on β (A, B, E) and α (C, D, F) cells as AVP.

(A, C) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B, D) (top panel) Time distribution of all measured events' halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 2 time periods where events per minute per ROI and AUC parameters were sampled for panels E and F. (bottom panel) Representative traces from ROIs indicated in A and C. (E) There were no significant differences in the pooled AUC data for β cells treated with the $[(D\text{-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$ and forskolin or with forskolin only. Direct comparison of the AUC values before and after the ligand application shows a large heterogeneity of responses. (F) A significant increase in the events per minute per ROIs due to $[(D\text{-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$ administration in α cells was noted. (G) There was no significant effect on insulin release from slices after $[(D\text{-Leu})^2, \text{Ile}^3, \text{Thr}^4]\text{-VP}$ exposure in comparison to stimulatory forskolin only. (H, I) The $d[\text{Cha}^4, \text{Dab}^8]\text{-VP}$ (a specific V_{1b} receptor agonist) and AVP produced similarly heterogeneous responses in β cells with non-significant changes in the AUC compared to forskolin only. All the parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

and oxytocin receptors, it is plausible that the stimulatory effect through V_{1b} receptors has been attenuated. Analyzing the effects of $d[Cha^4, Dab^8]$ -VP or AVP under similar experimental conditions, revealed that both produced comparably heterogeneous responses; however, paired analysis indicated a significant reduction in AUC with both V_{1b} receptor-specific agonists (**Fig. 6FG**) as well AVP (**Fig. 6I**).

Subsequently, we investigated the activation of V_{1A} receptors exclusively using $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]$ -VP, a V_{1A} receptor antagonist. As shown this antagonist did not inhibit AVP-stimulated Ca^{2+} activity in either α and β cells (**Fig. 7**). Nonetheless, we assessed the test the V_{1A} antagonistic action of $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]$ -VP on smooth muscle activity in the blood vessels adjacent to the islets in the same tissue slice (**Fig. 7G**). AVP application increased the frequency of Ca^{2+} oscillations in smooth muscle cells, and this activity has been specifically inhibited by $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]$ -VP and restored when antagonist has been washed out, suggesting AVP activity was specifically through V_{1A} receptor (**Fig. 7E**). On the other hand, the application of V_2 receptor antagonist tolvaptan did not have any significant effect on α and β cells or any other cell type in the slice and could not reverse any of the AVP effects (Suppl. Fig. 4).

Analysis of the pharmacological modulation of V_{1b} receptors in β cells

Fig. 8 summarizes the analysis of the pharmacological modulation of AVP receptors in β cells within pancreatic tissue slices. The amplitude and the direction of the changes in the rate of Ca^{2+} oscillations can vary significantly following the application of different concentrations of AVP during the concentration ramp experiment (**Fig. 8A**). The initial activation observed at lower pM concentrations was followed by a relatively stable response near the peak of a bell-shaped dependence curve, which subsequently transitioned to inhibition at concentrations approaching or exceeding nM levels of AVP. The scattering of the rate of changes at different AVP concentrations resembles that obtained during static AVP incubation experiment (**Fig. 8B**), showing a range of responses: activation (46%), no effect (27%), and inhibition (17%) after applying AVP, $[(D-Leu)^2, Ile^3, Thr^4]$ -VP (**Fig. 8C**) or $d[Cha^4, Dab^8]$ -VP (**Fig. 8D**). In contrast, control without agonists (**Fig. 8E**) or the use of antagonists for non-expressing AVP receptors following AVP stimulation, did not significantly affect the relative rate change (**Fig. 8FG**). One interpretation of the observed variability with V_{1b} receptor activation is that different islets, during their plateau phase stimulated by 8 mM glucose and forskolin, may achieve varying levels of IP_3 receptor activation (IP_3 receptor activation equivalent), before being further influenced by AVP or specific V_{1b} receptor agonists (**Fig. 8H**). Overall, the effect of V_{1b} receptor stimulation depends on the current functional status of an islet; resting β cells permit activation and increase insulin secretion, whereas in highly activated β cells, AVP and V_{1b} receptor agonists tend to promote inactivation and interfere with insulin release.

Discussion

The use of freshly prepared pancreatic tissue slices proved essential in unravelling the delicate role that AVP receptor signaling plays in the physiology of pancreatic β and α cells. This is a mouse model of normal, healthy living pancreatic tissue that is acutely removed from the animal and imaged in real time using high-resolution confocal microscopy shortly after extraction from the animal. We use minimal tissue manipulation during slicing and the vital Ca^{2+} indicator loading. Currently, this is the best in vivo tissue proxy that preserves all the properties of intact cells, among which it is crucial to highlight the preservation of intercellular connections and

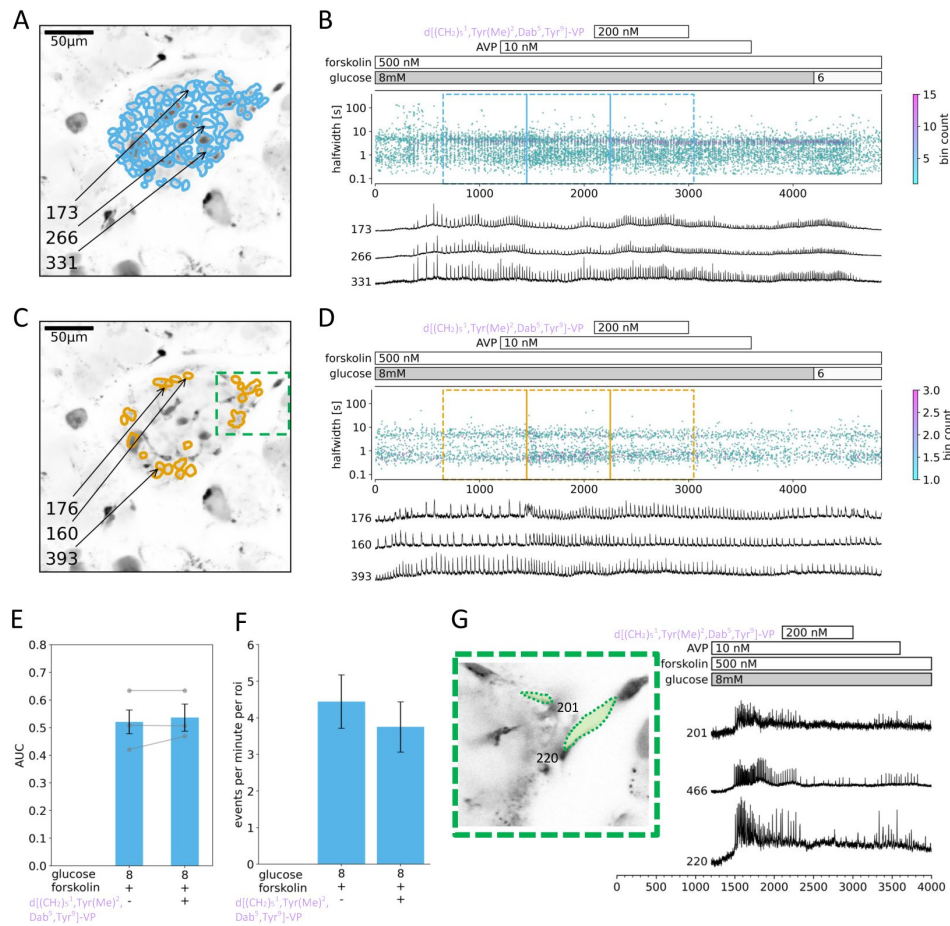


Figure 7.

A specific V_{1a} receptor antagonist has no significant effect on β (A, B, E, F) and α (C, D) cells, but inhibits activity in smooth muscle cells

(G). (A, C) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. Inset labelled with green dashed line is expanded in panel G. (B, D) (top panel) Time distribution of all measured events halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 3 time periods where events per minute per ROI and AUC parameters were sampled for panels E and F. (bottom panel) Representative traces from ROIs indicated in A and C. (E) There were no significant differences in the pooled AUC data for β cells treated with the $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]-VP$ and forskolin or with forskolin only. Direct comparison of the AUC values before and after the antagonist application shows no difference in responses. (F) No significant differences in the events per minute per ROIs due to $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]-VP$ administration in α cells were noted. (G) (left) Inset from C, showing location of two smooth muscle cells lining the blood vessel adjacent to the islet (right) Representative traces from ROIs on smooth muscle cells exposed to stimulatory glucose, forskolin, AVP, and $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]-VP$. AVP administration increased the frequency of Ca^{2+} oscillations in smooth muscle cells, but V_{1a} receptor antagonist fully, but reversibly inhibited this activity. All the parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD.

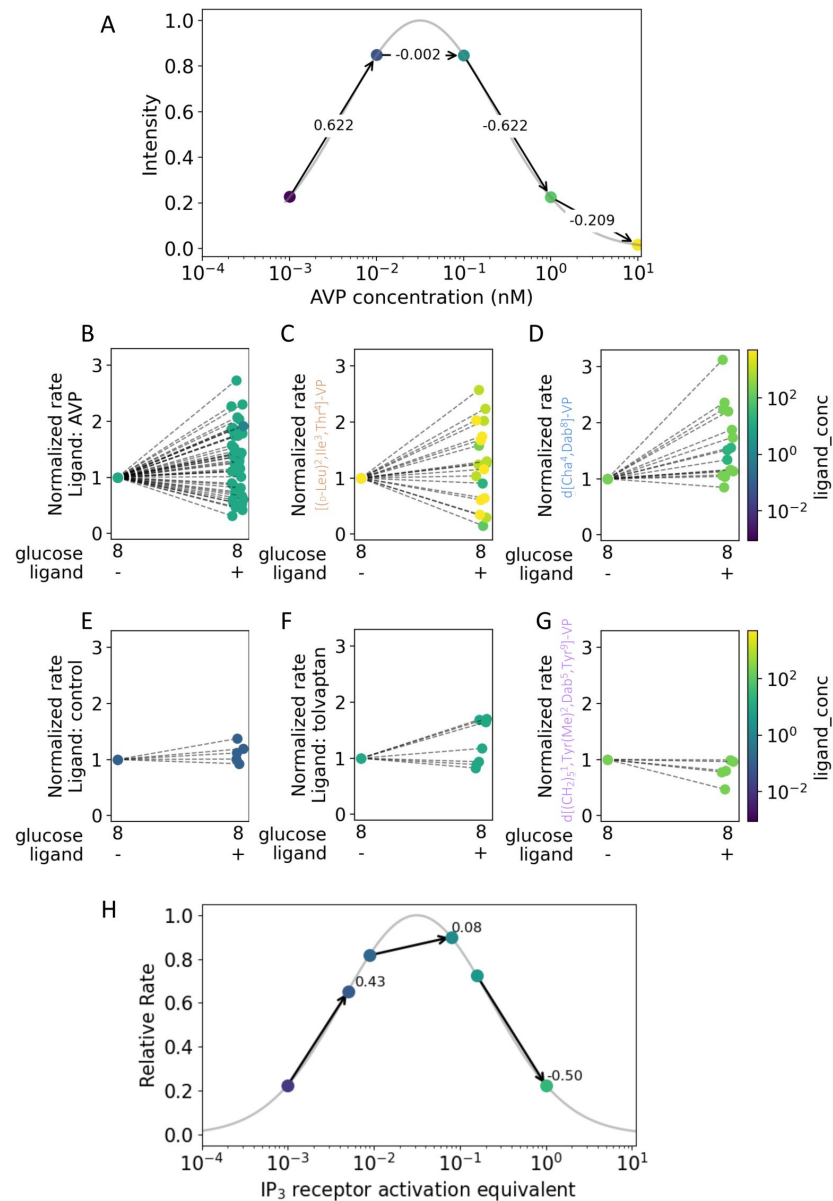


Figure 8.

Overview of the analysis of pharmacological modulation of V_{1b} receptors in β cells in pancreatic tissue slices.

(A) The relative rates of Ca^{2+} oscillations obtained during the AVP concentration ramp experiment. The bell-shaped dependence curve rises at the lower pM range of AVP, followed by a relatively stable response around the peak of the curve, and then falls at higher concentrations (nM range) of AVP. (B, C, D) Normalized rates to the basal stimulatory conditions (8 mM glucose) in the context of single concentration static incubation with AVP, $[(D-Leu)^2, Ile^3, Thr^4]-VP$ and $d[Cha^4, Dab^8]-VP$ are widely scattered. (E, F, G) Normalized rates to the basal stimulatory conditions (8 mM glucose) in the context of single concentration static incubation without AVP receptor agonists (control), with tolaptan (V_{1a} antagonist) and $d[(CH_2)_5^1, Tyr(Me)^2, Dab^5, Tyr^9]-VP$ (V_{1a} antagonist) are only slightly scattered. (H) Different islets during their plateau phase stimulated by 8 mM glucose and forskolin can achieve different levels of IP_3 receptor activation equivalent, contributing to the heterogeneity of further IP_3R stimulation with AVP or specific V_{1b} agonists.

communication between homotypic and heterotypic cell types. This communication is disrupted in research models involving dispersed islets (52) and cell lines. Our model also avoids the enzymatic treatment of tissue used for islet isolation (38, 53).

The existing evidence about the role of AVP for the activity of pancreatic β and α cells is controversial. Opinions vary widely, with some studies suggesting that AVP plays a significant role in releasing insulin and glucagon, while others indicate minimal effects (7, 14). There has been more evidence for functional expression of the V_{1b} receptor than regarding its mRNA or protein abundance in β cells. A single-cell transcriptome analysis (54) or protein expression analysis (8) of islets reported the absence of AVP receptors in β cells. On the other hand, overexpression of V_{1b} receptors in β cells improved the function of transplanted pseudoislets in vitro (55). Comparing cellular studies with intact tissue or in vivo studies is problematic for several reasons. First, the isolation and cultivation of islets and cell cultures require prolonged incubation in different culturing media, usually containing high glucose concentrations, exhausts β cells and decreases insulin secretion (56). Second, using different enzymatic and non-enzymatic cell dispersion agents interrupts plasma membrane proteins, such as receptors and gap-junctional proteins, responsible for intracellular communication and the synchronized response of islets to various stimuli (57–60). After the cell dispersal and disruption of intercellular connections, the expression and functionality of surface receptors can be significantly altered (61). Third, several sub-types of β cells have been identified and shown to play different roles in the dynamics of β cell collective response to stimulatory glucose (62). Last but not least, in different physiological and pathological settings, transcription of specific genes in β cells changes (63, 64), which may also happen during the preparation and culturing of cell lines (65). Future improvements in cellular approaches to assess different levels of expression, as well as physiological approaches like ours, are needed to resolve this issue, and to assess the existence of interactions between α cells, which show mRNA and protein expression, and β cells that do not.

In contrast to cellular and isolated islet studies, here our research aimed to reassess the role of AVP in the function of both cell types in a fresh pancreas tissue slice preparation, which has shown to be a preparation with in vivo-like sensitivity to physiological and pharmacological stimuli while at the same time enabling high spatial and temporal resolution measurements (38, 45, 47, 66). We focused on the G_q -receptor coupled V_{1b} receptor and its subsequent downstream signaling pathways involving IP_3 Rs. The bell-shaped response curve observed with AVP receptor agonists, where lower concentrations stimulate and higher concentrations inhibit cell activation, aligns with the biphasic activation and inactivation properties of the IP_3 receptors (37, 65, 67). The single concentration agonist experiments with an incubation time of 15 minutes allowed us to exclude the possibility of time-dependent effects that could occur during the prolonged AVP ramp experiments and lead to receptor internalization and desensitization. This bell-shaped relationship has significant implications for understanding how β and α cells, respond to fluctuating levels of AVP, particularly under varying plasma glucose levels, plasma osmolality and hormonal context. Importantly, it underscores the necessity of controlling AVP to avoid dysfunctional endocrine responses, or improve it after transplantation (55).

Since hormones acting via IP_3 as a Ca^{2+} -releasing messenger induce intracellular Ca^{2+} oscillations, which are influenced by Ca^{2+} feedback on the IP_3 receptor (68), we employed a combination of high-resolution measurements of cytosolic Ca^{2+} oscillations and a high-throughput analysis of specific newly synthesized V_{1b} receptor ligands, which simultaneously antagonized V_{1a} and oxytocin receptors. The findings describe a novel mechanism on how AVP modulates the activity of β and α cells, highlighting a permissive role of AVP that depends on increased cAMP levels due to concomitant activation of G_s -coupled hormonal receptors. The observed effects across the physiological AVP concentration range with IP_3 receptor signaling as a functional readout, followed a bell-shaped activation/inactivation profile for both cell types. The described non-linear functional dependence suggests that several controversies in previous studies may be due to the

fact that the experimental conditions were not sufficiently controlled to draw the appropriate conclusions, or that the experiments were interpreted without taking into account the apparent activation/inactivation profile previously observed in perfused rat pancreas (19) and insulinoma cells (29). By measuring the AVP-dependent activity of both cell types simultaneously, our results do not support the mechanism that the V_{1b} receptor activation in α cells could indirectly activate β cells in a paracrine fashion, but rather that activation of β and α cells is independent. Furthermore, a qualitatively similar bell-shaped dependence to acetylcholine stimulation has been shown for β cells, although interpreted differently (69).

Glucose-Dependent Modulation by AVP

Our results clearly reproduced the glucose-dependent effects of AVP on both α and β cells, corroborating previous studies. One of the differences with the majority of other studies is that AVP had its peak activity in its physiological osmoregulatory range of 10 to 100 pM (1) and not in the nM range as in other studies. AVP augmented β cell activity required stimulatory glucose concentration. AVP treatment has been able to activate α cells even at glucose-concentration that are typically inhibitory for these cells and as long as cAMP levels were elevated by forskolin directly stimulating adenylyl cyclase activity or physiologically with epinephrine stimulating G_s protein-coupled β adrenergic receptors. Furthermore, simultaneous α and β cell activity, enabled by the presence of forskolin, showed no major α - β cell-cell interaction at the time scales observed in this study, β cell being more sensitive to AVP, peaking between 10 and 100 pM, while the activity of α cells peaked at 1 nM AVP. It remains to be investigated whether the apparent lack of α - β cell communication is due to the fact that AVP primarily drives Ca^{2+} release through IP_3 receptors and less through plasma membrane-related Ca^{2+} fluxes.

Even a moderate supraphysiological glucose concentration shifted the AVP concentration dependence of β cells to the left, such that any physiologically relevant concentration of AVP could only have an inhibitory effect on the activity of these cells. This suggests that in hyperglycemia and increased plasma osmolality the rise in AVP concentration with concomitant elevation of cytosolic cAMP concentration through GLP-1 or other G_s -protein-coupled receptor stimulation, could effectively diminish the capacity of β cells to adequately activate or can even lead to suppression of glucose-dependent insulin release. Similarly, a glucose-dependent modulation of AVP-dependency could have a significant effect on the activation capacity of α cells in hypoglycemic conditions and contribute to the inability of α cells to counteract hypoglycemia. The observed right shift in the AVP-dependence of α cells in comparison to β cells could be the result of a generally lower level of activation due to suboptimal glucose concentration for the activity of α cells and could be sensitized at a more favorable glucose concentration. Further experiments will help clarifying this important issue. It is important to note that while AVP can potentiate α cell activity, this potentiation has been found to be inadequate in T1D patients to support the counter-regulatory response during hypoglycemia, a context that has been found to be associated with high-systemic concentration of AVP (14). The overactivation of α cells by AVP in the presence of high cAMP levels, due to the concomitant presence of stress-related adrenergic stimulation, could potentially desensitize or impair the α cells' ability to respond appropriately to dangerously low glucose conditions. This implies that under certain pathological or extreme physiological conditions, the presence of AVP, coupled with elevated cAMP, could disrupt normal glucagon release, compromising the body's ability to effectively counteract hypoglycemia.

Permissive Role of AVP in cAMP-Dependent Signaling

A critical aspect of AVP's role in β and α cells is its dependency on cAMP levels. Our study provides evidence that AVP's effects were contingent on the intracellular cAMP environment, with forskolin potentiating AVP's impact on both cell types at the same time, despite differential sensitivity to glucose concentration. This dependence suggests that AVP does not act as a primary activator of β and α cells, but rather serves to enhance the response of these cells to other stimuli when cAMP levels are elevated. The permissive role of AVP, particularly in the context of cAMP-dependent

signaling, aligns with the broader understanding of its function in other endocrine systems, such as CRH- or histamine-dependent potentiation of ACTH release in the pituitary (34). The observation that epinephrine, which reduces cAMP levels in β cells, suppresses AVP's effects further supports the notion that AVP's action is closely tied to the cAMP signaling axis. This permissive role may help explain the variability in AVP's effects observed across different studies and species, where varying experimental conditions could alter intracellular cAMP levels, thereby modulating AVP's impact. Due to its permissive effects, it is also not surprising that the antagonists of GLP-1 receptors prevent the AVP-dependent action in β cells (54).

V_{1b} Receptor-Specific Modulation

The heterogeneous response observed across different islets, where some showed increased activity while others did not, suggests that there may be variability in V_{1b} receptor expression or signaling capacity among β cells. This variability could be due to differences in receptor density, coupling efficiency to G_q proteins, or downstream signaling components such as PLC or IP₃Rs. The pharmacological profile of AVP, as well as the newly synthesized V_{1b} receptor-specific probes, supports the idea that V_{1b} receptors play an important but complex role in modulating pancreatic endocrine function.

The study specifically targeted the V_{1b} receptor to isolate its role from other AVP receptor subtypes that have been previously described to be expressed in the islet cells, including V_{1a}, V₂ and oxytocin receptors. The use of selective V_{1b} receptor agonists and V_{1a} and OTR antagonists allowed for a more precise characterization of the V_{1b} receptor's role in β and α cell function. The findings confirmed that V_{1b} receptors are the primary mediators of AVP's effects on these cells, with no detectable contribution from V_{1a}, V₂ or oxytocin receptors in the context of physiological glucose stimulation in pancreatic tissue slices. In addition, previous functional studies using selective V_{1b} receptor agonists, such as d[Cha4]AVP, revealed nanomolar affinity for V_{1b} receptors across species and concentration-dependent activation of natural biological models expressing these receptors (70).

Physiological Implications and Broader Context

Maintaining AVP levels within a physiological range is particularly challenging in the context of diabetes mellitus. The condition often leads to hyperosmolarity due to elevated glucose levels, plasma volume depletion due to increased water loss, both of which can stimulate AVP release or alternatively due to a direct stimulation of AVP release due to hypoglycemia (14). In such scenarios, the regulation of AVP becomes even more critical, as elevated AVP can exacerbate the dysregulation of α and β cell functions.

Additionally, the complex interplay between AVP, glucose levels, and the endocrine response in diabetes highlights a potential therapeutic challenge. On one hand, AVP's permissive role in cAMP-mediated signaling suggests that modulating AVP or its receptor pathways could fine-tune pancreatic islet responses under certain conditions. On the other hand, the risk of desensitization or inappropriate activation of α and β cells due to fluctuating AVP levels poses a significant challenge, particularly in diabetic patients who experience frequent shifts between hyperglycemia and hypoglycemia. The bell-shaped response curve of AVP, where low concentrations are stimulatory and high concentrations are inhibitory, further complicates the story. In a diabetic state, where AVP levels could be persistently elevated due to chronic hyperosmolarity and dehydration or hypoglycemia, the likelihood of tipping this balance into the IP₃R inhibitory range increases. This would not only impair insulin secretion but could also exacerbate glycemic instability by blunting the glucagon response during hypoglycemia.

Given these complexities, it is evident that managing AVP levels and its receptor signaling in diabetes requires a nuanced approach. Therapeutic strategies might need to consider not just glucose and insulin levels but also the broader neurohormonal environment, including AVP, to

ensure stable glycemic control. Future research could explore targeted modulation of AVP receptors, particularly V_{1b} receptors, as means of optimizing islet function without inducing the adverse effects associated with hyperosmolality-induced AVP elevation.

In summary, while AVP plays a crucial role in modulating pancreatic endocrine function, its regulation becomes particularly challenging in the context of diabetes mellitus. The difficulty in maintaining AVP within a physiological range due to plasma hyperosmolality and dehydration or hypoglycemia underscores the need for careful consideration of AVP signaling in managing diabetes, particularly in preventing hypoglycemic episodes and ensuring proper glucagon response.

Data availability

Raw data supporting the peptide synthesis & analysis, as well as pharmacological and physiological measurements of this study are available from the corresponding authors upon reasonable request.

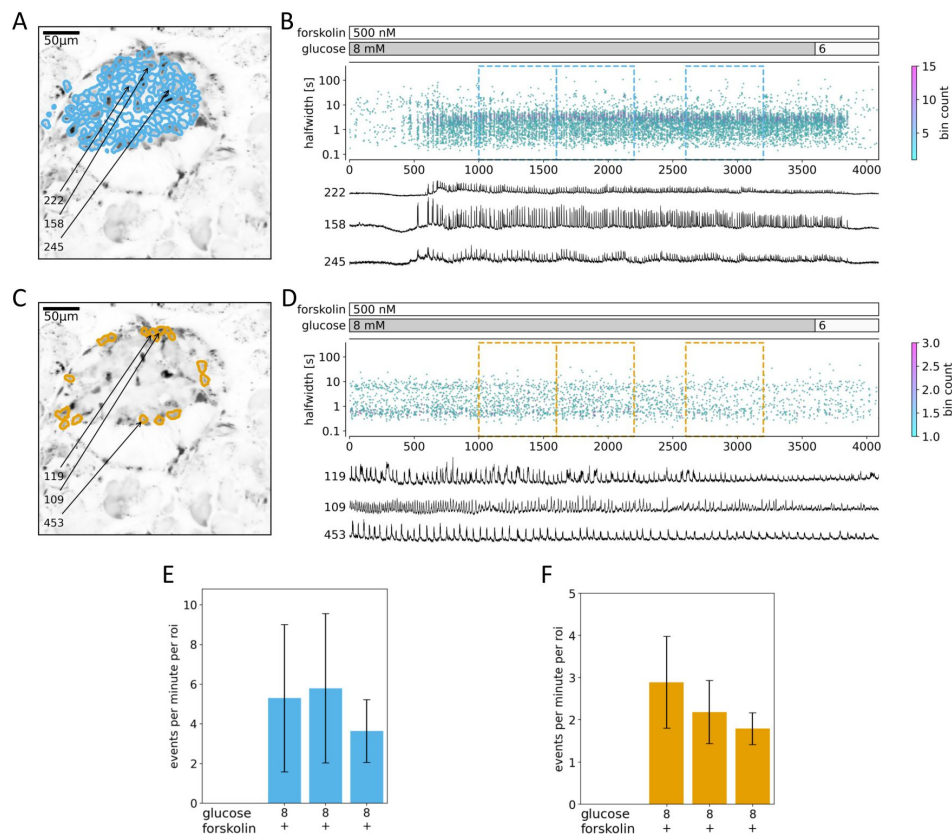
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Additional information

Author Contributions

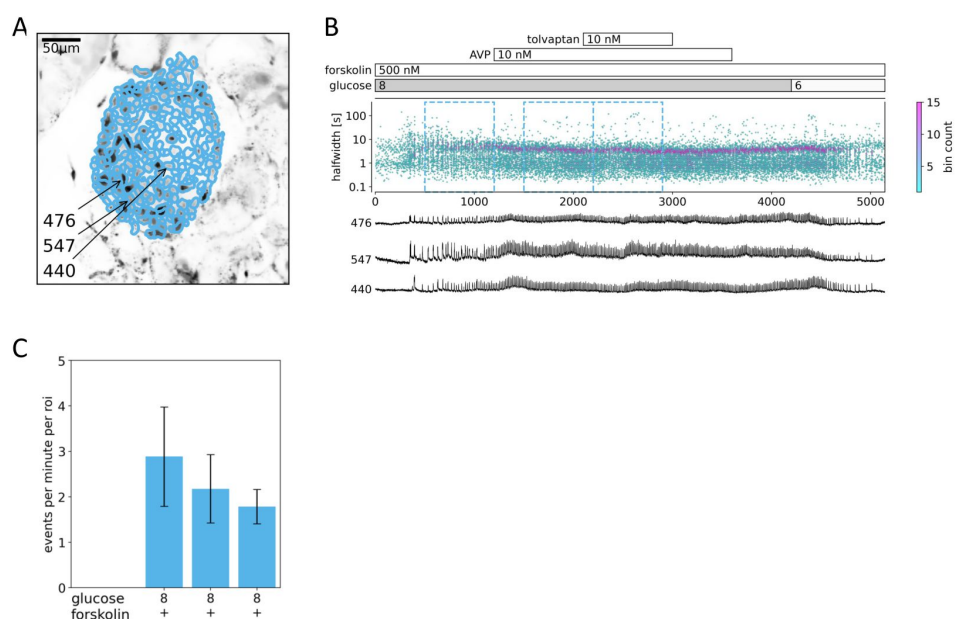
J. K., N. M., L. K. B., E. P. L., J. P., S. P. performed pancreatic tissue slice imaging experiments. N. M. and J. P. performed insulin release assay. M. S. R. and J. P. analyzed the data and created the figures. M. S. R., J. K., L. K. B., C. W. G., drafted the paper and all authors edited the paper. C. W. G., M. P., M. M., and X. K. provided resources. M. S. R., A. S., D. K., C. W. G., and M. M. acquired funding.



Supplementary Figure 1

The effect of physiological levels of glucose and forskolin on β (A, B, E, G, H) and α (C, D, F) cells.

(A, C) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B, D) (top panel) Time distribution of all measured events' halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 3 time periods where events per minute per ROI parameter was sampled for panels E and F. (bottom panel) Representative traces from ROIs indicated in A and C. (E, F) Prolonged exposure to physiological glucose stimulation and low forskolin concentration resulted in a stable Ca^{2+} oscillations over almost an hour in β and α cells. The parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD.



Supplementary Figure 2

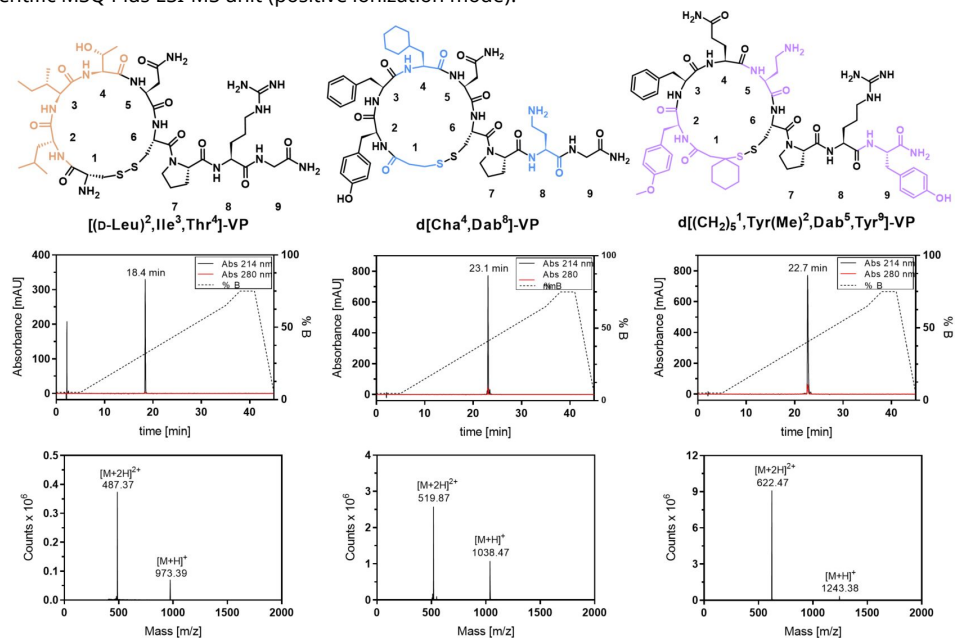
Tolvaptan, a V_2 receptor antagonist, has no significant effect on β cells.

(A) Indicated are the ROIs whose filtered traces correlate best with the average trace for the whole islet. (B) (top panel) Time distribution of all measured events' halfwidths at their peak times. Stimulation protocol is indicated above. The color bar indicates the bin count in each time/halfwidth point. The dashed-line rectangles indicate the 3 time periods where events per minute per ROI parameter was sampled for the panel. (bottom panel) Representative traces from ROIs indicated in A (C) Tolvaptan had no significant effects on the number of events per minute per ROI. The parameters were distributed normally, we used one-way ANOVA and Tukey post-hoc test. The data are presented as mean $\pm$ SD.

Supplementary Figure 3

Overview of molecular structures, chromatograms, and mass spectra of synthesized compounds.

Amino acid residues that differ from native vasopressin were marked in different colors. Analytic RP-HPLC chromatograms were measured for product purity determination at 214 nm. Solvent A (ddH₂O + 0.1% TFA) and B (ACN + 0.08% TFA) were used as eluents at 1 mL/min flow rate and a linear gradient of 5-65% B in 30 min on a Thermo Fisher Scientific Vanquish Horizon UHPLC system using a Kromasil Classic C₁₈ column (4.6 x 150 mm, 300 Å, 5 µm). Mass spectra were measured for product identity verification through direct injection on a Thermo Scientific Dionex Ultimate 3000 system equipped with a Thermo Scientific MSQ Plus ESI-MS unit (positive ionization mode).



Supplementary Table 1

Name, sequence, purity, and mass identity of synthesized vasopressin analogs and vasopressin.

Amino acid residues that differ from native vasopressin were marked in different colors.

Name	Sequence	Purity	m/z (calc.)	m/z (obs.)
Vasopressin (AVP)	CYFQNCPRG	—	—	—
[(D-Leu) ² ,Ile ³ ,Thr ⁴]-VP	C ^I I ^T NCPRG*	>99%	973.47 [M+H] ⁺	973.39 [M+H] ⁺
d[Cha ⁴ ,Dab ⁸]-VP	dCYF(Cha)NCP(Dab)G*	92.3%	1038.45 [M+H] ⁺	1038.47 [M+H] ⁺
d[(CH ₂) ₅ ¹ ,Tyr(Me) ² ,Dab ⁵ ,Tyr ⁹]-VP	d(CH ₂) ₅ Y ^{Me} FQ(Dab)CPRY*	98.3%	1243.58 [M+H] ⁺	1243.38 [M+H] ⁺

Cha = cyclohexylalanine; (CH₂)₅ = β-mercapto-β,β-cyclopentamethylenepropionic acid; d = deaminated N-terminus; Dab = 2,4-diaminobutyric acid; * = C-terminal amide.

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Reviewer #1 (Public review):

Summary:

The authors confirmed earlier findings that AVP influences α and β cells differently, depending on glucose concentrations. At substimulatory glucose levels, AVP combined with forskolin - an activator of cAMP - did not significantly stimulate β cells, although it did activate α cells. Once glucose was raised to stimulatory levels, β cells became active, and α cell activity declined, indicating glucose's suppressive effect on α cells and permissive effect on β cells. Under physiological glucose levels (8-9 mM), forskolin enhanced β -cell calcium oscillations, and AVP further modulated this activity. However, AVP's effect on β cells was variable across islets and did not significantly alter AUC measurements (a combined indicator of oscillation frequency and duration). In α cells, forskolin and AVP led to increased activity even at high glucose levels, suggesting that α cells remain responsive despite expected suppression by insulin and glucose.

Experiments with physiological concentrations of epinephrine suggest that AVP does not operate via Gs-coupled V2 receptors in β cells, as AVP could not counteract epinephrine's inhibitory effects. Instead, epinephrine reduced β cell activity while increasing α cell activity through different G-protein-coupled mechanisms. These results emphasize that AVP can potentiate α -cell activation and has a nuanced, context-dependent effect on β cells.

The most robust activation of both α and β cells by AVP occurred within its physiological osmo-regulatory range (~10-100 pM), confirming that AVP exerts bell-shaped concentration-dependent effects on β cells. At low concentrations, AVP increased β cell calcium oscillation frequency and reduced "halfwidths"; high concentrations eventually suppressed β cell activity, mimicking the muscarinic signaling. In α cells, higher AVP concentrations were

required for peak activation, which was not blunted by receptor inactivation within physiological ranges.

Attempting to further dissect the role of specific AVP receptors, the authors designed and tested peptide ligands selective for V1b receptors. These included a selective V1b agonist; a V1b agonist with antagonist properties at V1a and oxytocin receptors; and a selective V1a antagonist. In pancreatic slices, these peptides seem to replicate AVP's effects on Ca^{2+} signaling, although responses were highly variable, with some islets showing increased activity and others no change or suppression. The variability was partly attributed to islet-specific baseline activity, and the authors conclude that AVP and V1b receptor agonists can modulate β cell activity in a state-dependent manner, stimulating insulin secretion in quiescent cells and inhibiting it in already active cells.

Strengths:

Overall, the study is technically advanced and provides useful pharmacological tools. However, the conclusions are limited by a lack of direct mechanistic and functional data. Addressing these gaps through a combination of signaling pathway interrogation, functional hormone output, genetic validation, and receptor localization would strengthen the conclusions and reduce the current (interpretive) ambiguity.

Weaknesses:

- (1) The study is entirely based on pharmacological tools. Without genetic models, off-target effects or incomplete specificity of the peptides cannot be fully ruled out.
 - (2) Despite multiple claims about β cell activation or inhibition, the functional output - insulin secretion - is weakly assessed, and only in limited conditions. This aspect makes it very hard to correlate calcium dynamics with physiological outcomes.
 - (3) Insulin and glucagon secretion assays should be provided; the authors should measure hormone release in parallel with Ca^{2+} imaging, using perfusion assays, especially during AVP ramp and peptide ligand applications.
- Additionally, there is no standardization of the metabolic state of islets. The authors should consider measuring islet NAD(P)H autofluorescence or mitochondrial potential (e.g., using TMRE) to control for metabolic variability that may affect responsiveness.
- (4) There is a high degree of variability in response to AVP and V1b agonists across islets (activation, no effect, inhibition). Surprisingly, the authors do not fully explore the cause of this heterogeneity (whether it is due to receptor expression differences, metabolic state, experimental variability, or other conditions).
 - (5) There is no validation of V1b receptor expression at the protein or mRNA level in α or β cells using in situ hybridization, immunohistochemistry, or spatial transcriptomics.
 - (6) AVP effects are described in terms of permissive or antagonistic effects on cAMP (especially in relation to epinephrine), but direct measurements of cAMP in α and β cells are not shown, weakening these conclusions. The authors should use Epac-based cAMP FRET sensors in α and β cells to monitor the interaction between AVP, forskolin, and epinephrine more conclusively.
 - (7) Single-islet transcriptomics or proteomics (also to clarify variability) should be provided to analyze receptor expression variability across islets to correlate with response phenotypes (activation vs inhibition). Alternatively, the authors could perform calcium imaging with simultaneous insulin granule tracking or ATP levels to assess islet functional states.

(8) While the study implies AVP acts through V1b receptors on β cells, the signaling downstream (e.g., PLC activation, IP3R isoforms involved) is simply inferred but not directly shown.

(9) The interpretation that IP3R inactivation (mentioned in the title!) underlies the bell-shaped AVP effect is just hypothetical, without direct measurements. Assays in β (and/or α)-cell-specific V1b KO mice and IP3R KO mice must be provided to support these speculations.

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Reviewer #2 (Public review):

Summary:

In this paper, Drs. Kerčmar, Murko, and Bombek make a series of observations related to the role of AVP in pancreatic islets. They use the pancreatic slice preparation that their group is well known for. The observations on the slice physiology are technically impressive. However, I am not convinced by the conclusions of this manuscript for a number of reasons. At the core of my concern is perhaps that this manuscript appears to be motivated to resolve 'controversies' surrounding the actions of AVP on insulin and glucagon secretion. This manuscript adds more observations, but these do not move the field forward in improving or solidifying our mechanistic understanding of AVP actions on islets. A major claim in this manuscript is the beta cell expression of the V1b Receptor for AVP, but the evidence presented in this paper falls short of supporting this claim. Observations on the activation of calcium in alpha cells via V1b receptor align with prior observations of this effect.

I have focused my main concerns below. I hope the authors will consider these suggestions carefully - please be assured that they were made with the intent to support the authors and increase the impact of this work.

Strengths:

The main strength of this paper is the technical sophistication of the approach and the analysis and representation of the calcium traces from alpha and beta cells.

Weaknesses:

(1) The introduction is long and summarizes a substantive body of literature on AVP actions on insulin secretion in vivo. There are a number of possible explanations for these observations that do not directly target islet cells. If the goal is to resolve the mechanistic basis of AVP action on alpha and beta cells, the more limited number of papers that describe direct islet effects is more helpful. There are excellent data that indicate that the actions of AVP are mediated via V1bR on alpha cells and that V1bR is a) not expressed by beta cells and b) does not activate beta cell calcium at all at 10 nM - which is the same concentration used in this paper (Figure 4G) for peak alpha cell Ca^{2+} activation (see <https://doi.org/10.1016/j.cmet.2017.03.017>; cited as ref 30 in the current manuscript).

(2) We know from bulk RNAseq data on purified alpha, beta, and delta cells from both the Huising and Gribble groups that there is no expression of V2a. I will point you to the data from the Huising lab website published almost a decade ago (<http://dx.doi.org/10.1016/j.molmet.2016.04.007>) - which is publicly available and can be used to generate figures (<https://huisinglab.com/data-ghrelin-ucsc/index.html>). They indicate the absence of expression of not only AVP2 receptors anywhere in the islet, but also the lack of expression of V1bra, V1brb, and Oxtr in beta cells. Instead of the detailed list of expression of these 4 receptors elsewhere in the body, it would be more directly relevant to set up their pancreatic slice experiments to

summarize the known expression in pancreatic islets that is publicly available. It would also have helped ground the efforts that involved the generation of the V1aR agonist and V2R antagonist, which confirm these known AVP/OXT receptor expression patterns.

(3) Importantly, the lack of V1br from beta cells does not invalidate observations that AVP affects calcium in beta cells, but it does indicate that these effects are mediated a) indirectly, downstream of alpha cell V1br or b) via an unknown off-target mechanism (less likely). The different peak efficacies in Figure 4G would also suggest that they are not mediated by the same receptor.

(4) The rationale for the use of forskolin across almost all traces is unclear. It is motivated by a desire to 'study the AVP dependence of both alpha and beta cells at the same time'. As best as I can determine, the design choice to conduct all studies under sustained forskolin stimulation is related to the permissive actions of AVP on hormone secretion in response to cAMP-generating stimuli. The permissive actions by AVP that are cited are on hormone secretion, which in many cell types requires activation of both calcium and cAMP signaling. Whether the activation of V1br and subsequent calcium response is permitted by cAMP is unclear. I believe the argument the authors are making here is that the activation of beta cell calcium by AVP is permitted by forskolin. i.e., the cAMP stimulated by it in beta cells. However, the design does not account for the elevation of cAMP in alpha cells and subsequent release of glucagon, particularly upon co-stimulation with AVP, which permits glucagon release by activating a calcium response in alpha cells. This glucagon could then activate beta cells. If resolving the mechanism of action is the goal, often less is more. The activation of Gαq-mediated calcium is not cAMP dependent (although the downstream hormone secretion clearly often is). As was shown, AVP does not activate calcium in beta cells in the absence of cAMP. The experiments in Figures 1, 2, and 4 should have been completed in the absence of cAMP first.

(5) It is unexpected that epinephrine in Figure 2 does not activate the alpha cell calcium? A recent paper from the same group (Sluga et al) shows robust calcium activation in alpha cells in a similar prep by 1 nM epinephrine, which is similar to the dose used here.

(6) Figure 8 suggests a pharmacological activation of beta cell V1bR in the low pM range. How do the authors reconcile this comparison with the apparent absence of an effect of AVP stimulation at low pM to low nM doses in beta cells (Figure 4A)? I note that there are changes over time with sustained beta cell stimulation with 8 mM glucose, but these changes are relatively subtle, gradual, and quite likely represent the progression of calcium behaviors that would have occurred under sustained glucose, irrespective of these very low AVP concentrations. I will note that the K_d of the V1bR for AVP is around 1 nM, with tracer displacement starting around 100 pM according to the data in figure 5B, which is hard to reconcile with changes in beta cell calcium by AVP doses that start 10-100-fold lower than this dose at 1 and 10 pM (Figure 8).

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Reviewer #3 (Public review):

Summary:

This work aims to better understand the role of arginine vasopressin (AVP) in the control of islet hormone secretion. This builds on previous literature in this area reporting on the actions of AVP to stimulate islet hormones. The gap in literature being addressed by these studies is primarily focused on the glucose-dependency of AVP on both insulin and glucagon secretion. A secondary objective is to explore the role of individual receptors with the use of newly generated peptides and existing tools. The methods include the use of Ca²⁺ imaging in

pancreas slices from mice, with additional outcomes including insulin secretion in some areas. The conclusions presented are that AVP acts through V1b receptors in both alpha- and beta-cells, that this activity occurs in the high cAMP environment, and is glucose dependent.

Strengths:

The area of research is emerging with plenty of room for new contributions. The concept of AVP stimulating islet hormone secretion is important and deserving of further insight. The use of pancreas tissue to image primary cells makes the experiments physiologically relevant. The advancement of novel tools in this area should be helpful to other groups investigating the actions of AVP.

Weaknesses:

The conclusions are only modestly supported by the data and lack experimental depth and rigor. The rationale for only conducting studies at high cAMP conditions is not entirely clear and limits the conclusions that can be made. The use of Ca²⁺ is helpful, but it is a surrogate for hormone secretion. Additional measurements of hormone secretion are needed to enhance the robustness of these conclusions. Consideration of paracrine effects between alpha- and beta-cells is only superficially made and is likely essential in the context of the experimental design. For instance, there is clear literature that alpha-cells secrete several factors that work in paracrine interactions on beta-cells and autocrine actions back on alpha-cells. Conducting these studies in a high cAMP context only completely overlooks these interactions, skewing the interpretations made by the investigators. Finally, the clarity of the experiments and results could be significantly enhanced.

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Author response:

We sincerely thank the reviewers and editors for their thoughtful and constructive evaluation of our manuscript and their recognition of its technical strengths, including advanced spatio-temporal Ca²⁺ imaging, image processing, and the rational design of selective AVP receptor ligands. We appreciate their acknowledgement that our study contributes to the understanding of glucose-dependent AVP effects in pancreatic islet physiology. Their comments will guide us to refine the scope of our work, which focuses on how α and β cells respond to AVP under varying glucose and hormonal conditions, rather than on linear correlations between the function and transcript levels in individual cells or metabolic profiles in individual cell. Most of the reviewers' concerns and proposed remedies reflect a reductionist framework, for which we believe cannot not fully account for emergent behavior within the islet collective. As we and others have shown, islet cells do not behave in isolation; their responses often depend on the state of the entire cell population(1, 2). This means that even under identical experimental conditions, responses can differ depending on the islet's current state. These patterns are not random, but reflect how the islet integrates signals dynamically(3, 4).

To take advantage of both the systems and molecular side, we do plan to address several of the reviewers' suggestions with new experiments and analyses:

First, we will add hormone, specifically glucagon, secretion assays to support our conclusions on α cell responses and possible paracrine effects. Second, we will include a targeted transcript analysis of V1bR using RNAscope and extend the pharmacological characterization of downstream signaling using selective agonists and inhibitors. Third, we will clarify the rationale for using forskolin, and added new experiments using GLP-1 analogues to selectively increase cAMP in β cells, allowing us to examine direct AVP effects. And fourth, we

will reinforce presence of emergency and that variability in islet responses is not experimental noise, but a hallmark of the collective, non-linear behavior of the islet cell collective, which should later drive a rethinking of experimental designs and the interpretation of pharmacological responses. In conclusion, we believe that our study provides new insights into AVP modulation in pancreatic islets and highlights the importance of context-dependent responses in α and β cells. We are grateful for the opportunity to revise our manuscript and look forward to further strengthening it further through the review process.

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